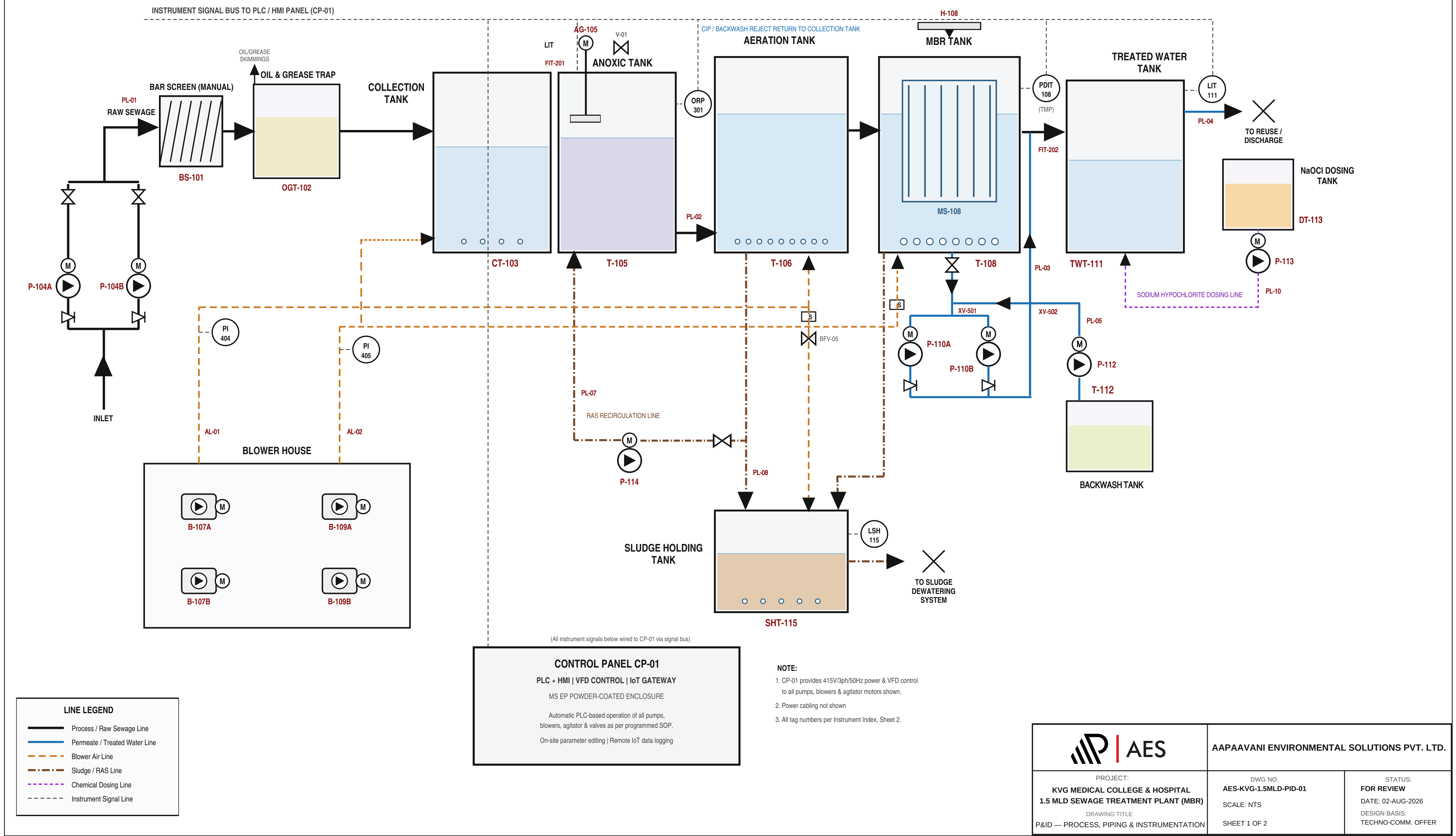


PIPING & INSTRUMENTATION DIAGRAM (P&ID)

1.5 MLD SEWAGE TREATMENT PLANT — MBR PROCESS — KVG MEDICAL COLLEGE & HOSPITAL



P&ID SUPPORTING SCHEDULES

EQUIPMENT LIST | INSTRUMENT INDEX | LINE LIST | GENERAL NOTES

1. EQUIPMENT LIST			
Tag No.	Equipment Description	Qty	Key Data / Specification
BS-101	Manual Bar Screen	1	Coarse screening, manually raked
OGT-102	Oil & Grease Trap	1	Gravity separation chamber
S-101	Screen	1	For pre-aeration duty
CT-103	Collection cum Equalization Tank	1	RCC tank, provides feed to biological section
P-104A/B	Feed Pump (Duty + Standby)	1+1	5 HP, self-priming, CI body / SS impeller, Kirloskar/CG
T-105	Anoxic Tank	1	RCC tank, denitrification zone
AG-105	Anoxic Agitator	1	Slow-speed submersible agitator, SS, Kirloskar/CG
T-106	Aeration Tank	1	RCC tank, fine bubble diffused aeration
T-108	MBR Tank	1	RCC tank housing submerged membrane skid
MS-108	Membrane Skid + Membrane Modules	1 Lot	SS304 skid; submerged hollow-fibre reinforced PVDF membrane, 0.05 µm pore, 40 m²/module, Make: Mitsubishi
H-108	Membrane Lifting Hoist	1	4 T electric chain hoist with pendant control, I-beam monorail
P-110A/B	Permeate Pump (Duty + Standby)	APD	3 HP, self-priming, CI body / SS impeller, Kirloskar/CG
TWT-111	Treated Water Tank	1	RCC tank, clear water storage prior to disinfection/reuse
T-112	Backwash Tank	1	For membrane backwash
P-112	Backwash Pump	1	2 HP, self-priming, CI body / SS impeller
DT-113	Disinfectant (NaOCl) Dosing Tank	1	HDPE dosing tank
P-113	Disinfectant Dosing Pump	APD	Metering / dosing pump
P-114	RAS Pump	1	2 HP sewage pump, CI/SS impeller, for sludge recirculation to Anoxic Tank
SHT-115	Sludge Holding Tank	1	RCC tank, aerobic sludge storage prior to screening/dewatering
—	Rotary Drum Screen	1	Design flow 100 m³/hr, screen opening 2-4 mm, MOC SS304
B-107A/B	Blower & Motor — Aeration Tank & Sludge Holding Tank (Duty+Standby)	1+1	1200 m³/hr @ 0.6 kg/cm², 40 HP / 1440 RPM, Make: Akash / CG or equiv.
B-109A/B	Blower & Motor — MBR Scouring & Collection Tank (Duty+Standby)	1+1	1200 m³/hr @ 0.6 kg/cm², 40 HP / 1440 RPM, Make: Akash / CG or equiv.
—	Fine Bubble Disc Diffuser	150 Nos	5-6 m³/hr each, 12" dia., PVC & EPDM, Make: WTE Italy
—	Coarse Bubble Disc Diffuser	60 Nos	10-15 m³/hr each, 5" dia., PVC & EPDM, Make: WTE Italy
XV-501/502	Electric Solenoid Valve (automatic valve control)	2 Nos	Make: Aira
CP-01	Control Panel	1	PLC + HMI, VFD control, IoT gateway, PLC Make: Mitsubishi, MSEP powder-coated enclosure

2. LINE TYPE & SYMBOL	
	Process / Raw Sewage Line
	Permeate / Treated Water Line
	Blower Air Line
	Sludge / RAS Line
	Chemical Dosing Line Instrument
	Signal Line (to CP-01)

Instrument bubbles: circle = field-mounted device, dashed-ring circle = analyzer (DO/ORP/TMP)

3. INSTRUMENT INDEX			
Tag No.	Service / Location	Instrument Type	Function / Signal
LIT-103	Collection Tank CT-103	Level Transmitter	Continuous level to CP-01, feed pump control
LSHH-103	Collection Tank CT-103	Level Switch (High-High)	Feed pump trip / high level alarm
FIT-201	Feed line, CT-103 to T-105	Electromagnetic Flow Meter	Influent flow indication, Make: Fluido Sense
ORP-301	Anoxic Tank T-105	ORP Analyzer	Process monitoring (good-practice addition)
-	-	-	-
PDIT-108	MBR Tank / Membrane Skid MS-108	Differential Pressure Transmitter	Trans-membrane pressure (TMP), fouling/CIP trigger
LIT-111	Treated Water Tank TWT-111	Level Transmitter	Continuous level to CP-01, permeate pump control
FIT-202	Permeate / Treated water outlet	Electromagnetic Flow Meter	Product flow indication, Make: Fluido Sense
LSH-115	Sludge Holding Tank SHT-115	Level Switch (High)	High level alarm
PI-402	Permeate pump discharge header	Pressure Gauge	Local indication
PI-404	Blower B-107 discharge header	Pressure Gauge	Local indication
PI-405	Blower B-109 discharge header	Pressure Gauge	Local indication
XV-501	Permeate line	Solenoid-Operated Butterfly Valve	Automatic valve control, Make: Aira
XV-502	Backwash line	Solenoid-Operated Butterfly Valve	Automatic valve control, Make: Aira

4. LINE LIST			
Line No.	Service	From / To	Size & Material
PL-01	Raw sewage	Inlet → BS-101 → OGT-102 → CT-103	UPVC, Make: Finolex/Astral
PL-02	Feed to bio-reactor	CT-103 → P-104A/B → T-105 → T-106 → T-108	UPVC
PL-03	Permeate (product water)	T-108 → P-110A/B → TWT-111	UPVC
PL-04	Treated water discharge	TWT-111 → disinfection → reuse/discharge	UPVC
PL-05	Backwash supply	T-112 → P-112 → MS-108	UPVC
PL-06	-	-	-
PL-07	RAS recirculation	T-106/T-108 → P-114 → T-105	UPVC
PL-08	Excess sludge	T-106/T-108 → SHT-115 → DS-116	UPVC
PL-09	-	-	-
PL-10	Chemical dosing (NaOCl)	DT-113 → P-113 → TWT-111 inlet	UPVC, chemical duty
AL-01	Blower air – Header A	B-107A/B → T-106 & SHT-115	GI/UPVC, 8"4"/2.5" main/branch/diffuser
AL-02	Blower air – Header B	B-109A/B → T-108 (scouring) & CT-103	GI/UPVC, 8"4"/2.5" main/branch/diffuser

5. GENERAL
1. Process basis: MBR (Membrane Bio-Reactor) technology; design capacity 1.5 MLD (1500 KLD). 2. All instrumentation shown is wired back to Control Panel CP-01 (PLC + HMI + IoT) via the instrument signal bus; refer Sheet 1 for routing and Item 2 above for tag details. 3. Electrical power distribution (415V/3ph/50Hz) from CP-01 / MCC to all motors is not shown on this P&ID. 4. Scope of supply includes supply, transportation, loading, installation and commissioning of all equipment, plus 20 kg bacterial culture for plant start-up and 50 litres of disinfectant chemical for initial trial run. 5. Blower Header-A serves Aeration Tank (T-106) & Sludge Holding Tank (SHT-115); Header-B serves MBR scouring (T-108) & Collection Tank (CT-103) mixing air, each via a dedicated 40 HP duty + 40 HP standby blower. 6. ORP-301 analyzers are recommended good-practice process instruments for MBR control loops. 7. All pipe work in UPVC (high grade) with brackets & plumbing, Make: Finolex/Astral, unless noted.

AES	AAPAAVANI ENVIRONMENTAL SOLUTIONS PVT. LTD.	
PROJECT: KVG MEDICAL COLLEGE & HOSPITAL 1.5 MLD SEWAGE TREATMENT PLANT (MBR) DRAWING TITLE P&ID SUPPORTING SCHEDULES	DWG NO. AES-KVG-1.5MLD-PID-01 SCALE: NTS SHEET 2 OF 2	STATUS: FOR REVIEW DATE: 02-AUG-2026 DESIGN BASIS: TECHNO-COMM. OFFER